Statistical Analysis of Road Accident Patterns and Severity Outcomes in Bangladesh

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Abstract

Road safety has become a critical issue due to rising injuries, fatalities, and economic losses, especially in Bangladesh where rapid urbanization and high vehicle density increase accident risks. This study uses statistical analysis on a large real-world accident dataset with both qualitative and quantitative variables. Data preprocessing, descriptive analysis, and visualization were applied to identify patterns and key factors affecting accident severity. The findings highlight the influence of road conditions, traffic characteristics, and vehicle-related factors, providing data-driven insights to support improved road safety and policy decisions.

Introduction

- Road safety is a growing global concern due to rising injuries, deaths, and economic losses.
- Bangladesh faces high accident risks because of rapid urbanization and heavy traffic.
- Poor traffic management and high vehicle density increase accident frequency.
- Road accidents have become a major socioeconomic issue in the country.
- Statistical analysis helps identify key factors behind accidents.
- This study uses a dataset of ~49,000 accident records.
- Both qualitative and quantitative variables are analyzed.
- The goal is to find patterns and support better road safety decisions.

Methodology

- **Data Source:** Real-world road traffic accident dataset (~49,000 records) with qualitative and quantitative variables.
- **Data Preprocessing:** Performed in Microsoft Excel by removing duplicates, checking unrealistic values, and detecting outliers using the IQR method.
- **Variables Used:** Accident intensity, driver age, vehicle count, injury count, death count, road types, and traffic control systems.
- **Qualitative Analysis:** Bar charts and pie charts for categorical data visualization.
- **Quantitative Analysis:** Histograms, scatter plots, and frequency curves for numerical data analysis.
- **Tools Used:** Microsoft Excel and Python for data analysis,visualization and tests..

Data Visuals

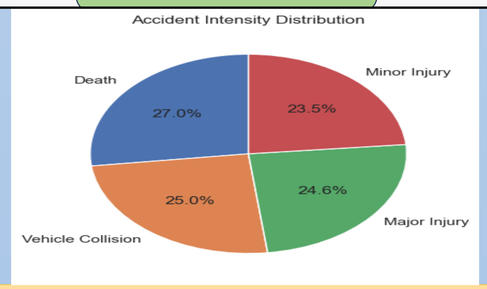


Figure 1: Road accident severity is a multi-factor problem, requiring broader systemic analysis rather than focusing on isolated variables.

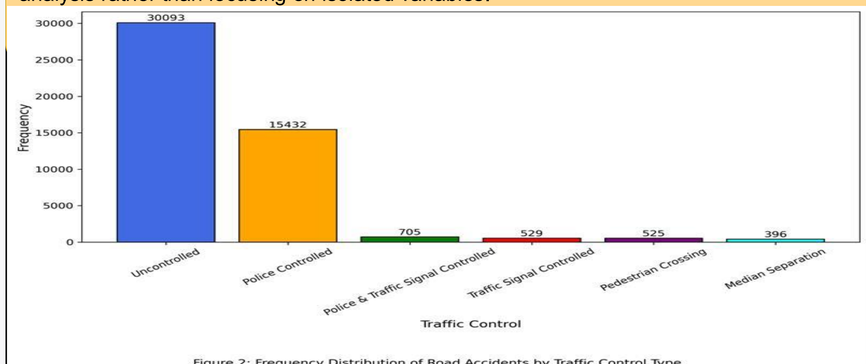


Figure 2: Frequency Distribution of Road Accidents by Traffic Control Type

Figure 2: The accidents occur more frequently at **uncontrolled intersections** than at controlled ones, suggesting that the lack of proper traffic regulation contributes to higher accident occurrence.

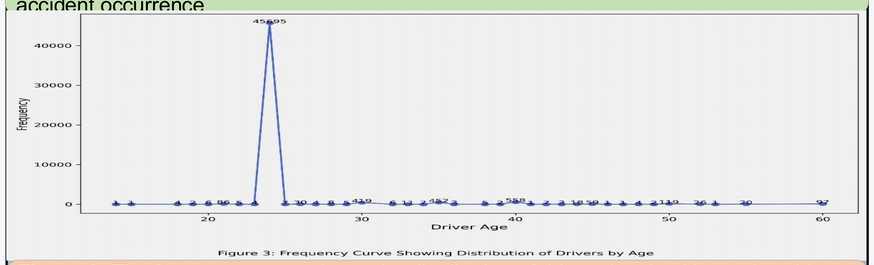


Figure 3: Frequency Curve Showing Distribution of Drivers by Age

Figure 3: The frequency curve shows the distribution of drivers involved in road accidents by age. It indicates that accidents occur most frequently among drivers in their **mid-twenties**. The frequency decreases for higher age groups.

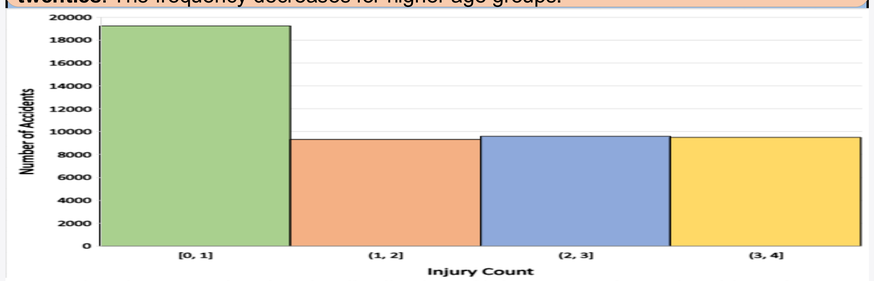


Fig: Histogram showing the distribution of injury counts in road accidents in Bangladesh.

Figure 4: Histogram of injury counts in road accidents in Bangladesh shows that most accidents result in 0–1 injuries, while higher injury counts (1–4) are less frequent and occur at similar low levels.

Results & Key Findings

Correlation Analysis

- Hypothesis: Injury count is related to accident severity.
- There is no significant relationship between injury count and accident severity.
- Accident severity is influenced by other factors, not just the number of injuries.

Independent T-Test

- Hypothesis: Accident severity differs between daylight and night.
- The test was inconclusive due to insufficient or imbalanced data.
- No reliable conclusion can be made about the effect of lighting conditions.

Chi-Square Test

- Hypothesis: Weather conditions affect accident risk levels.
- Weather and accident risk are independent.
- Weather alone is not a strong factor in determining accident risk.

Regression Analysis

- Purpose: To identify factors influencing accident risk.
- Vehicle count and fatalities significantly influence accident risk.
- Injury count does not have a significant effect.
- Accident risk is more dependent on traffic conditions and fatal outcomes rather than injury numbers.

Conclusion

Road accident severity in Bangladesh is driven by multiple factors like multi-vehicle crashes, uncontrolled intersections, driver age, and traffic conditions, showing the need for stronger traffic management and systemic analysis rather than isolated focus.